Md Ahsanul Haque, Ph.D.
MAT-120

Homework-4

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Instructions:

- Write your full name (first **and** last name) **inside** the provided box.
- This assignment contains 6 problems. It is **YOUR responsibility** to check that your copy has all 6 problems.
- Make sure to read the instructions for each problem carefully.
- Use correct mathematical notation.
- Write cohesively and follow a logical sequence of steps.
- Check that your answer can be read and understood by the grader.

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1. (10 points)
   (a) Check the integrability of the following functions on the given intervals. Justify your reasoning in each case:
      i. $f(x) = \begin{cases} 2x+1 & 0 \le x < 2 \\ 7-x & 2 \le x \le 4 \end{cases}$ on the interval $[0, 4]$.
      ii. $g(x) = \frac{1}{(x-2)^2}$ on the interval $[0, 3]$.
   (b) Evaluate the following indefinite integrals using an appropriate method:
      i. $\displaystyle\int \frac{x^3}{x^2+1}\,dx$
      ii. $\displaystyle\int \frac{1}{x\,(1+(\ln x)^2)}\,dx$
2. (10 points) Evaluate the following limits:
   (a) $\displaystyle\lim_{x\to 0} \frac{x - \sin x}{x^3}$
   (b) $\displaystyle\lim_{x\to\infty} \left(1 - \frac{3}{x}\right)^{2x}$
   (c) $\displaystyle\lim_{x\to 0^+} \left(\frac{1}{x} - \frac{1}{\sin x}\right)$
   (d) $\displaystyle\lim_{x\to 0} \frac{e^x - e^{-x} - 2x}{x - \sin x}$

3. (10 points) Consider the function $f(x) = x^2 e^{-x}$ defined on the domain $[0, \infty)$.
   (a) Find $f'(x)$ and determine the open intervals where $f(x)$ is strictly increasing and strictly decreasing. Find all relative (local) extrema.
   (b) Determine the open intervals of concavity and state any points of inflection.
   (c) Find the absolute maximum and absolute minimum values of $f(x)$ on the closed interval $[0, 5]$.
   (d) Evaluate $\lim_{x\to\infty} f(x)$. What does this limit imply about the absolute extrema of $f(x)$ on its full domain?

4. (10 points) Solve the following problems:
   (a) A particle moves along a straight line with position function $s(t) = t^3 - 6t^2 + 9t + 2$ for $t \geq 0$, where s is in meters and t is in seconds.
      i. Find the velocity $v(t)$ and acceleration $a(t)$ functions.
      ii. Determine the time interval(s) during which the particle is speeding up (moving faster) and slowing down.
   (b) An inverted conical water tank with vertex down has a top radius of 4 m and a height of 8 m. Water is pumped into the tank at a constant rate of 5 m^3/min, but leaks out from the bottom at 2 m^3/min.
      i. Express the volume V of water in the tank strictly as a function of the water depth h.
      ii. How fast is the water level rising at the exact instant when the water depth is 3 m?

5. (10 points) Evaluate the following definite integrals:
   (a) $\displaystyle\int_0^{\sqrt{3}} \frac{1}{1+x^2}\,dx$
   (b) $\displaystyle\int_0^4 |x-2|\,dx$
   (c) $\displaystyle\int_0^{\pi/2} \cos^2(x)\,dx$
   (d) $\displaystyle\int_1^e \frac{1}{x(1+\ln x)}\,dx$

6. (10 points) Solve the following:
   (a) Use the Fundamental Theorem of Calculus to find the derivative of the following:
      i. $H(x) = \displaystyle\int_2^{x^3} \sqrt{1+t^2}\,dt$
      ii. $G(x) = \displaystyle\int_{\sin x}^{\cos x} e^{t^2}\,dt$
   (b) Determine the right-endpoint Riemann sum approximation for $f(x) = 2x^2 + 1$ on the interval $[1, 4]$ using $n = 6$ subintervals of equal width.
   (c) Express the following sums using sigma notation:
      i. $2^2 + 4^2 + 6^2 + \cdots + 20^2$
      ii. $\frac{1}{2} - \frac{2}{3} + \frac{3}{4} - \frac{4}{5} + \cdots + \frac{9}{10}$

Solution to question no. -1

1.a.i) Given: $f(x) = \begin{cases} 2x+1, & 0 \le x \le 2 \\ 7-x & 2 \le x \le 4 \end{cases}$

interval : $[0,4]$

at $x=2$,

$f(2) = 7-2 = 5$

$\lim_{x \to 2^-} f(x) = \lim_{x \to 2^-} 2x+1$

$= 2 \times 2 + 1$

$= 5$

$\lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} 7-x$

$= 7-2$

$= 5$

$\therefore \lim_{x \to 2^-} f(x) = f(2) = \lim_{x \to 2^+} f(x) = 5$

$\therefore f(x)$ is continuous at $x=2$

$\therefore f(x)$ is integrable on $[0,4]$.

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1.a.ii) $g(x) = \frac{1}{(x-2)^2}$

interval : $[0, 3]$

$(x-2)^2 \neq 0$

$\therefore x \neq 2$

for, $x = 2$ $g(x)$ is undefined

$\therefore$ Vertical asymptote at $x = 2$

$\therefore g(x) = \frac{1}{(x-2)^2}$ is not integrable on $[0, 3]$

1.b.i) $\int \frac{x^3}{x^2+1} dx$

$= \int (x - \frac{x}{x^2+1}) dx$

$= \int x\, dx - \int \frac{x}{x^2+1} dx$

$= \frac{x^2}{2} - \int \frac{1}{2} \frac{1}{u} du$

$= \frac{x^2}{2} - \frac{1}{2} \ln u$

$= \frac{x^2}{2} - \frac{1}{2} \ln (x^2+1) + C$

let,

$u = x^2 + 1$

$du = 2x\, dx$

$\Rightarrow \frac{1}{2} du = x\, dx$

1.b.-ii) $\int \frac{1}{x(1+(\ln x)^2)}\, dx$

let,
$u = \ln x$
$du = \frac{1}{x} dx$

$= \int \frac{1}{1+u^2}\, du$

$= \tan^{-1} u$

$= \tan^{-1} \ln x + c$

Solution to question no. -2

a) $\lim_{x \to 0} \frac{x - \sin x}{x^3}$; $\frac{0}{0}$

$= \lim_{x \to 0} \frac{1 - \cos x}{3x^2}$; $\frac{0}{0}$

$= \lim_{x \to 0} \frac{\sin x}{6x}$

$= \frac{1}{6}$

b) $\lim_{x \to \infty} \left(1 - \frac{3}{x}\right)^{2x}$

Let, $y = \lim_{x \to \infty} \left(1 - \frac{3}{x}\right)^{2x}$

$\Rightarrow \ln y = \lim_{x \to \infty} \ln \left(1 - \frac{3}{x}\right)^{2x}$

$\Rightarrow \ln y = \lim_{x \to \infty} 2x \ln \left(1 - \frac{3}{x}\right)$

$\Rightarrow \ln y = \lim_{x \to \infty} \frac{2 \ln \left(1 - \frac{3}{x}\right)}{1/x}$; $\frac{0}{0}$

$\Rightarrow \ln y = \lim_{x \to \infty} \frac{2 \cdot \frac{1}{1 - \frac{3}{x}} \cdot \frac{3}{x^2}}{-\frac{1}{x^2}}$

$\Rightarrow \ln y = \lim_{x \to \infty} \frac{-6}{1 - \frac{3}{x}}$

$\Rightarrow \ln y = -6$

$\therefore y = e^{-6}$

c) $\lim_{x \to 0^+} \left(\frac{1}{x} - \frac{1}{\sin x} \right)$

$= \lim_{x \to 0^+} \frac{\sin x - x}{x \sin x} \quad ; \frac{0}{0}$

$= \lim_{x \to 0^+} \frac{\cos x - 1}{\sin x + x \cos x} \quad ; \frac{0}{0}$

$= \lim_{x \to 0^+} \frac{-\sin x}{\cos x + \cos x - x \sin x}$

$= \frac{-0}{1+1-0}$

$= 0$

d) $\lim_{x \to 0} \frac{e^x - e^{-x} - 2x}{x - \sin x} \quad ; \frac{0}{0}$

$= \lim_{x \to 0} \frac{e^x + e^{-x} - 2}{1 - \cos x} \quad ; \frac{0}{0}$

$= \lim_{x \to 0} \frac{e^x - e^{-x}}{\sin x} \quad ; \frac{0}{0}$

$= \lim_{x \to 0} \frac{e^x + e^{-x}}{\cos x}$

$= \frac{1+1}{1}$

$= 2$

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Solution to question no. 3

a) $f(x) = x^2 e^{-x}$; $[0, \infty)$

$f'(x) = 2xe^{-x} - x^2 e^{-x}$

$= xe^{-x}(2-x)$

let, $xe^{-x}(2-x) = 0$

$\Rightarrow x = 0$ or, $2-x = 0$; $e^{-x} > 0$

$\Rightarrow x = 2$

$\therefore$ critical point, $x = 0, 2$

interval :



$[0, 2) \Rightarrow f'(1) = \frac{1}{e}$ $(+Ve)$ increasing

$(2, \infty) \Rightarrow f'(3) = \frac{-3}{e^3}$ $(-Ve)$ decreasing

$\therefore$ increasing on $[0, 2)$ and decreasing on $(2, \infty)$

Local minimum : $f(0) = 0$

Local maximum : $f(2) = \frac{4}{e^2}$

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b) We have, $f'(x) = e^{-x} x e^{-x}(2 - x^2)$

$= e^{-x}(2x - x^2)$

$f''(x) = e^{-x}(2 - 2x) - e^{-x}(2x - x^2)$

$= e^{-x}(x^2 - 4x + 2)$

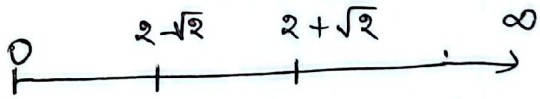
let,

$e^{-x}(x^2 - 4x + 2) = 0$

$e^{-x} \neq 0 \quad \therefore x^2 - 4x + 2 = 0$

$\Rightarrow x = \frac{4 \pm \sqrt{16 - 8}}{2}$

$\therefore x = 2 + \sqrt{2}, 2 - \sqrt{2}$



$[0, 2-\sqrt{2})$: $f''(\frac{1}{2}) = 0.15$; concave up

$(2-\sqrt{2}, 2+\sqrt{2})$: $f''(2) = -0.27$; concave down

$(2+\sqrt{2}, \infty)$: $f''(4) = 0.037$; concave up

concavity changes at : $2-\sqrt{2}$ and $2+\sqrt{2}$

$\therefore$ inflection points are: $(2-\sqrt{2}, (2-\sqrt{2})^2 e^{-(2-\sqrt{2})})$

and, $(2+\sqrt{2}, (2+\sqrt{2})^2 e^{-(2+\sqrt{2})})$

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c) Critical points we get from a,

$x = 0, 2$

For interval $[0, 5]$

$f(0) = 0$

$f(2) = \frac{4}{e^2} = 0.54$

$f(5) = \frac{25}{e^5} = 0.17$

∴ Absolute maximum $= \frac{4}{e^2}$ at $x = 2$

Absolute minimum $= 0$ at $x = 0$

d) $\lim_{x \to \infty} x^2 e^{-x}$

$= \lim_{x \to \infty} \frac{x^2}{e^x}$; $\frac{\infty}{\infty}$

$= \lim_{x \to \infty} \frac{2x}{e^x}$

$= \lim_{x \to \infty} \frac{2}{e^x}$

$= 0$

∴ As x approaches to ∞, the function approaches to 0.

∴ Absolute minimum remains on $x = 0$

and Absolute maximum at $x = 2$

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Solution to question no. 4

a) $s(t) = t^3 - 6t^2 + 9t + 2$

i) $v(t) = s'(t) = 3t^2 - 12t + 9$

$a(t) = v'(t)$

$= s''(t)$

$= 6t - 12$

ii) we get, $v(t) = 3t^2 - 12t + 9$

$= 3(t^2 - 4t + 3)$

$= 3(t-3)(t-1)$

$3(t-3)(t-1) = 0$

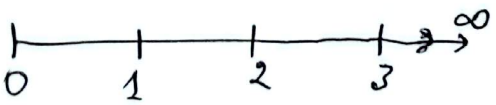
$\therefore t = 1, 3$

$a(t) = 6t - 12 = 6(t-2)$

$6(t-2) = 0$

$\therefore t = 2$

$\therefore$ critical points are $t = 1, 2, 3$



speeding up when v, a have same sign.

$(0, 1) \Rightarrow v(\frac{1}{2}) = 3.75$
$a(\frac{1}{2}) = -9$ > slowing down

$(1, 2) \Rightarrow v(\frac{3}{2}) = -\frac{9}{4}$
$a(\frac{3}{2}) = -3$ > speeding up

$(2,3) \Rightarrow v\left(\frac{5}{2}\right) = -\frac{9}{4}$ > slowing down

$a\left(\frac{5}{2}\right) = 3$

$(3,\infty) \Rightarrow v(4) = 9$ > speeding up

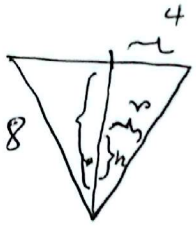
$a(4) = 12$

$\therefore$ Speeding up on $(1,2) \cup (3,\infty)$

Slowing down on $(0,1) \cup (2,3)$

b) i) $r = 4$ m

$h = 8$ m



$\frac{dV}{dt} = 5 - 2 = 3$ m^3/min

$\frac{r}{h} = \frac{4}{8}$

$\therefore r = \frac{h}{2}$

$V = \frac{1}{3}\pi r^2 h$

$= \frac{1}{3}\pi\left(\frac{h}{2}\right)^2 h$

$= \frac{\pi h^3}{12}$

ii) $V = \frac{\pi h^3}{12}$

$$\frac{dV}{dt} = \frac{\pi}{12} \times 3h^2 \frac{dh}{dt}$$

$$\Rightarrow 3 = \frac{\pi}{4} \times h^2 \frac{dh}{dt} \qquad \left[\because \frac{dV}{dt} = 3 \text{ m}^3/\text{min}\right]$$

$$\Rightarrow \frac{dh}{dt} = \frac{12}{\pi \times 3^2} \quad dt, \; h = 3$$

$$\therefore \frac{dh}{dt} = \frac{12}{9\pi} = \frac{4}{3\pi} \text{ m/min}$$

Solution to question no. 5

a) $\int_0^{\sqrt{3}} \frac{1}{1+x^2} dx$

$= \tan^{-1} x \Big|_{x=0}^{x=\sqrt{3}}$

$= \tan^{-1} \sqrt{3} - \tan^{-1} 0$

$= \frac{\pi}{3}$

b) $\int_0^4 |x-2|\,dx$

for, $0 \le x < 2$ $\quad |x-2| = 2-x$

for, $2 < x \le 4$ $\quad |x-2| = x-2$

$$\therefore \int_0^4 |x-2|\,dx = \int_0^2 2-x\,dx + \int_2^4 x-2\,dx$$

$$= 2x - \frac{x^2}{2}\Big|_{x=0}^{x=2} + \frac{x^2}{2} - 2x\Big|_{x=2}^{x=4}$$

$$= 2\times 2 - \frac{4}{2} + (8-8) - (2-4)$$

$$= 2+2$$

$$= 4$$

c) $\int_0^{\frac{\pi}{2}} \cos^2(x)\,dx$

$$= \int_0^{\frac{\pi}{2}} \frac{1+\cos 2x}{2}\,dx$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} 1+\cos 2x\,dx$$

$$= \frac{1}{2}\left[x + \frac{\sin 2x}{2}\right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2}\left(\frac{\pi}{2} + 0 - 0\right)$$

$$= \frac{\pi}{4}$$

d) $\int_1^e \frac{1}{x(1+\ln x)}\, dx$

let, $u = 1 + \ln x$

$du = \frac{1}{x}\, dx$

$x = 1 ;\ u = 1 + \ln 1$

$= 1$

$x = e ;\ u = 1 + \ln e$

$= 2$

$= \int_1^2 \frac{1}{u}\, du$

$= \ln u \Big|_{u=1}^{u=2}$

$= \ln 2 - \ln 1$

$= \ln 2$

Solution to question no.-6

a) i) $H(x) = \int_2^{x^3} \sqrt{1+t^2}\, dt$

FTC : $\frac{d}{dx} \int_a^{g(x)} f(t)\, dt = f(g(x))\, g'(x)$

$g(x) = x^3$

$g'(x) = 3x^2$

$\therefore H'(x) = \sqrt{1+(x^3)^2} \cdot 3x^2 = 3x^2 \sqrt{1+x^6}$

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ii) $G(x) = \int_{\sin x}^{\cos x} e^{t^2}\, dt$

$$\frac{d}{dx}\int_{u(x)}^{v(x)} f(t)\, dt = f(v(x))\, v'(x) - f(u(x))\, u'(x)$$

$v(x) = \cos x \quad ; \quad v'(x) = -\sin x$

$u(x) = \sin x \quad ; \quad u'(x) = \cos x$

$$\therefore G'(x) = e^{(\cos x)^2}(-\sin x) - e^{(\sin x)^2}(\cos x)$$

$$= -\sin x\, e^{\cos^2 x} - \cos x\, e^{\sin^2 x}$$

b) $f(x) = 2x^2 + 1$

interval $[1,4]$, $n = 6$

$$\Delta x = \frac{4-1}{6} = \frac{1}{2}$$

Subinterval : $[1, 1.5], [1.5, 2], [2, 2.5], [2.5, 3], [3, 3.5], [3.5, 4]$

$\therefore$ Right endpoint: $R_6 = \Delta x\{f(1.5) + f(2) + f(2.5) + f(3) + f(3.5) + f(4)\}$

$$= \frac{1}{2}(5.5 + 9 + 13.5 + 19 + 25.5 + 33)$$

$$= 52.75$$

c) i) $2^2+4^2+6^2+\ldots+20^2$

$= 2^2+(2\cdot 2)^2+(2\cdot 3)^2+\ldots\ldots+(2\cdot 10)^2$

$\therefore \sum_{k=1}^{10} (2k)^2$

ii) $\frac{1}{2}-\frac{2}{3}+\frac{3}{4}-\frac{4}{5}+\ldots\ldots+\frac{9}{10}$

$\frac{(-1)^2}{2}+\frac{2(-1)^3}{3}+\frac{3(-1)^4}{4}+\frac{4(-1)^5}{5}+\ldots-+\frac{9(-1)^{10}}{10}$

$\therefore \sum_{k=1}^{9} (-1)^{k+1}$